



**Schaltschrank-
Kühlgerät**

Cooling unit

Climatiseur

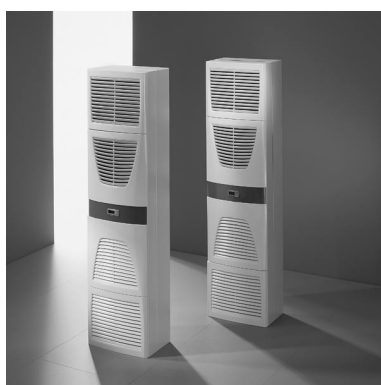
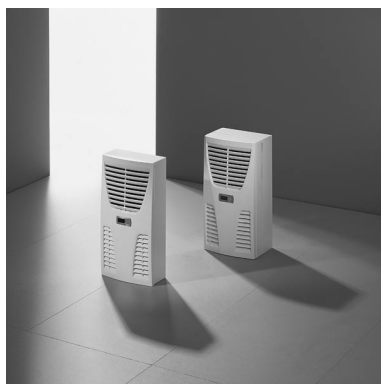
Koelaggregaat

Kylaggregat

**Condizionatore
per armadi**

**Refrigerador
para armarios**

**エンクロージャー用
クーリングユニット**



SK 3302.xxx SK 3304.xxx SK 3328.xxx SK 3332.xxx
SK 3303.xxx SK 3305.xxx SK 3329.xxx

Montageanleitung

Assembly instructions

Notice de montage

Montage-instructie

Montageanvisning

Istruzioni di montaggio

Instrucciones de montaje

取扱説明書

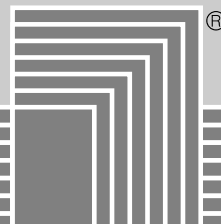


Abb. 3.1: SK 3302.xxx

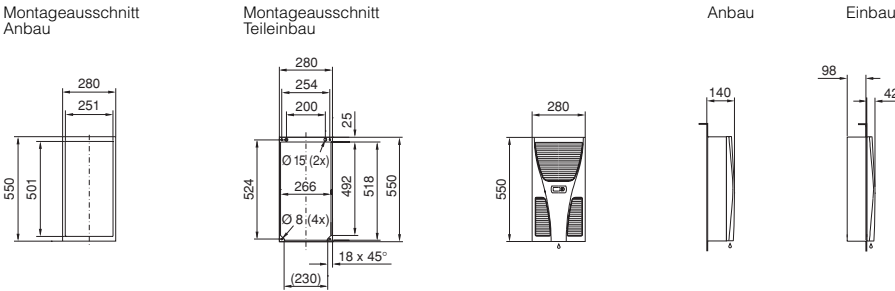


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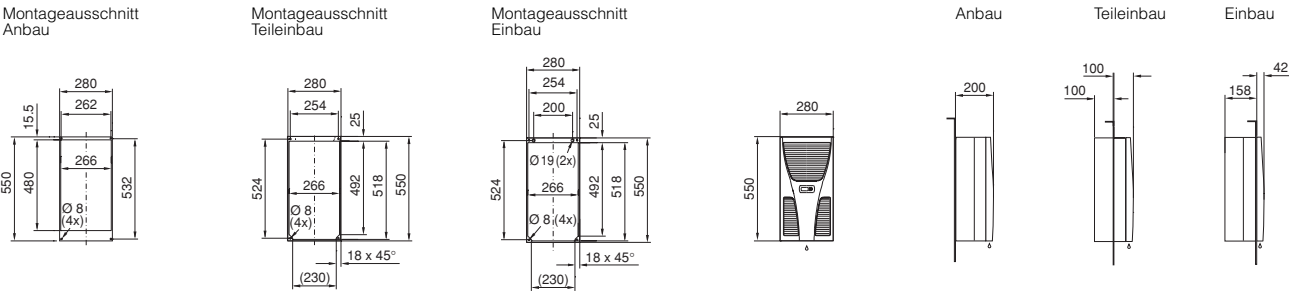


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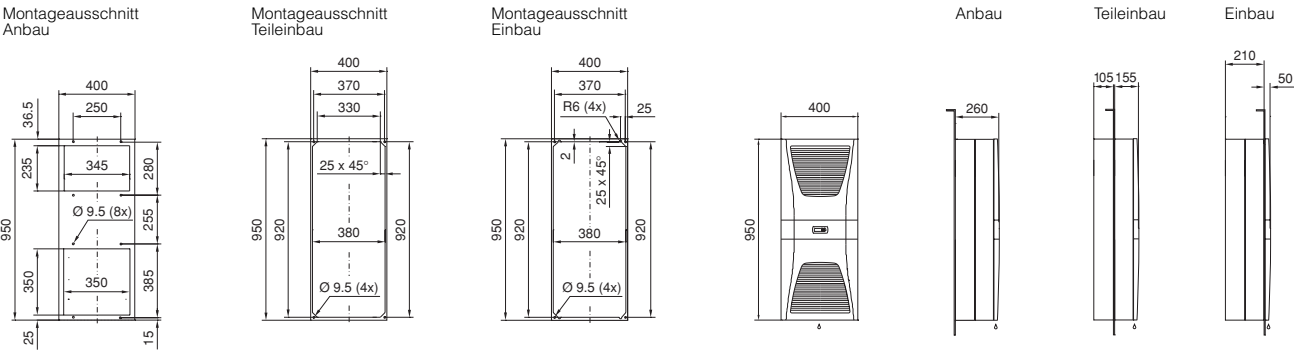


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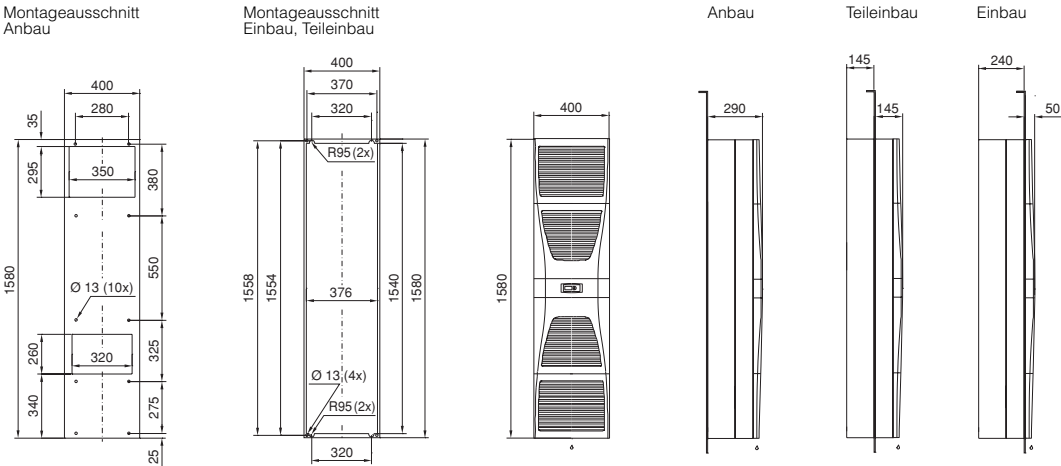


Abb. 3.2 Gerätemontage

Fig. 3.2 Mounting

Fig. 3.2 Montage de l'appareil

Afb. 3.2 Apparaatmontage

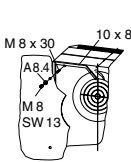
Bild 3.2 Aggregatmontage

Fig. 3.2 Montaggio dell'apparecchio

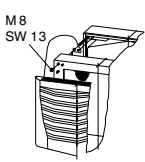
Fig. 3.2 Montaje del aparato

図3.2 取付けカットアウト図

Anbau
External
installation
En saillie
Aanbouw
Uitopbouw
Montaggio
sporgente
Montaje
exterior
表面取付け



Teileinbau
Partially internal
installation
Partiellment
intégré
Gedeeltelijke
inbouw
Delvis inbyggnad
Montaggio
semincassato
Montaje
parcial
半埋め込み取付け



Einbau
Internal
installation
Intégré
Inbouw
Inbyggnad
Montaggio
incassato
Montaje
interior
全埋め込み取付け

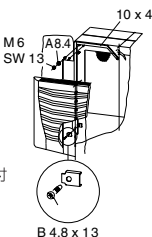
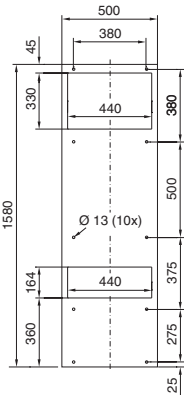
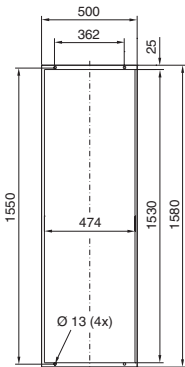


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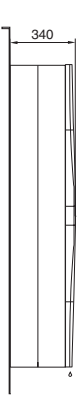
Montageausschnitt
Anbau



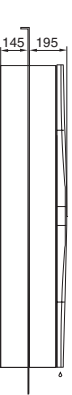
Montageausschnitt
Teileinbau



Anbau



Teileinbau



Tab. 2.1 Technische Daten

Tab. 2.1 Technical data

Tab. 2.1 Données techniques

Tab. 2.1 Technische gegevens

Tab. 2.1 Tekniska data

Tab. 2.1 Caratteristiche tecniche

Tab. 2.1 Datos técnicos

表2.1 仕様

D	Farbton RAL 7035	Edelstahlhaube 1.4301	Bemessungs- spannung	Bemes- sungs- strom	Anlauf- strom	Vor- siche- rung T	Einschalt- dauer	Bemessungs- leistung	Nutzkühlleistung	Kältemittel	zul. Druck PS	Temperat- urbereich	Geräusch- pegel	Schutzart Innenkreislauf Außenkreislauf	Abmessungen (B x H x T) mm	Gewicht
GB	Colour RAL 7035	Stainless steel cover 1.4301	Operating voltage	Rated current	Starting current	Pre-fuse T	Duty cycle	Nom. refrigeration	Useful cooling output	Refrigerant	Permis- sible pressure	Temperature range	Noise level	Protection categ. Internal circuit External circuit	Dimensions (W x H x D) mm	Weight
F	Coloris RAL 7035	Capot en acier inoxydable 1.4301	Tension nominale	Courant nominal	Courant de démarrage	Dispositif de sécurité T	Durée de mise en circuit	Puissance nominale	Puissance frigorigène de régime	Fluide frigori- gène	Pression de régime autor.	Plage de température	Niveau sonore	Degré de protect. Circuit intérieur Circuit extérieur	Dimensions (L x H x P) mm	Poids
NL	Kleur RAL 7035	RVS mantel 1.4301	Bedrijfs- spanning	Nominale stroom	Aanloop- stroom	Primaire zekering T	Inschakel- duur	Nominaal vermogen	Nuttig koelvermogen	Koel- middel	p. max.	Temperatuur- bereik	Geluids- nivo	Beschermklasse Inwendig circuit Uitwend. circuit	Afmetingen (B x H x D) mm	Gewicht
S	Färgton RAL 7035	Huv i rostfritt stål 1.4301	Anslutnings- spänning	Märk- ström	Startström	För- säkring gl	Ingångs- tid	Märkeffekt	Effektiv kyleffekt	Kylmedel	Tillåtet drifts- övertryck	Temperatur- område	Ljudnivå	Kapslingsklass Inre kretslopp Yttre kretslopp	Mått (B x H x D) mm	Vikt
I	Colore RAL 7035	Cover in acciaio inox 1.4301	Tensione nominale	Corrente nominale	Corrente di spunto	Fusibili T	Ciclo d'in- serzione	Potenza nominale	Potenza frigorifera utile	Fluido frigorigeno	Pressione max.	Campo di temperatura	Livello di rumore	Grado di protez. Circuito interno Circuito esterno	Dimensioni (L x A x P) mm	Peso
E	Color RAL 7035	Cubierta de acero inoxidable 1.4301	Tensión de servicio	Intensidad nominal	Intensidad de arranque	Fusible T	Duración de conexión	Potencia nominal	Potencia frigorífica útil	Fluido frigorífico	Presión máxima admis.	Campo de temperaturas	Nivel de ruido	Protección Circuito interior Circuito exterior	Dimensiones (anch. x alt. x prof.) mm	Peso
J	塗装色 RAL 7035	ステンレス製 カバー 1.4301	定格電圧	定格電流	始動電流	プリヒューズ	負荷サイクル	定格電力	有効冷却能力	冷媒	最大圧力	使用温度範囲	ノイズレベル	保護等級 内部循環 外部循環	外形寸法 幅×高さ×奥行き	質量
								L35 L35 L35 L50	DIN 3168/EN 814 L35 L35 L35 L50					EN 60 529		
SK 3302.110 SK 3302.100	SK 3302.210 SK 3302.200	115 V, 60 Hz 230 V, 50/60 Hz	3.3 A 1.6 A/ 1.7 A	8.0 A 3.0 A/ 3.4 A	10.0 A/ 10.0 A	100 %		290 W/340 W 245 W/255 W 255 W/275 W	300 W/320 W 150 W/170 W	R134a, 100 g R134a, 100 g	25 bar	+20 – +55°C	< 61dB (A)	IP 54/IP 34	280 x 550 x 140	13 kg
SK 3303.100 SK 3303.500	SK 3303.200 SK 3303.600	230 V, 50/60 Hz	2.6 A/ 2.6 A	5.1 A/ 6.4 A	10.0 A/ 10.0 A	100 %		360 W/380 W 420 W/390 W	500 W/610 W 280 W/350 W	R134a, 170 g	28 bar	+20 – +55°C	< 61dB (A)	IP 54/IP 34	280 x 550 x 200	17 kg
SK 3303.110 SK 3303.510	SK 3303.210 SK 3303.610	115 V, 60 Hz	5.7 A	11.5 A	10.0 A	100 %		470 W 500 W								
SK 3304.100 SK 3304.500	SK 3304.200 SK 3304.600	230 V, 50/60 Hz	4.8 A/ 4.4 A	12.0 A/ 14.0 A	10.0 A/ 10.0 A	100 %		700 W/650 W 750 W/710 W								39 kg
SK 3304.110 SK 3304.510	SK 3304.210 SK 3304.610	115 V, 50/60 Hz	9.5 A/ 10.0 A	26.0 A/ 28.0 A	16.0 A/ 16.0 A	100 %		725 W/680 W 780 W/750 W	1000 W/1060 W 790 W/ 840 W	R134a, 500 g	25 bar	+20 – +55°C	< 64 dB (A)	IP 54/IP 34	400 x 950 x 260	44 kg
SK 3304.140 SK 3304.540	SK 3304.240 SK 3304.640	400 V, 3 – 50 Hz/ 460 V, 3 – 60 Hz	2.5 A/ 2.6 A	11.5 A/ 12.7 A	10.0 A*/ 10.0 A*	100 %		580 W/550 W 660 W/680 W								40 kg
SK 3305.100 SK 3305.500	SK 3305.200 SK 3305.600	230 V, 50/60 Hz	5.4 A/ 6.0 A	22.0 A/ 24.0 A	16.0 A/ 16.0 A	100 %		850 W/1000 W 1000 W/1160 W								41 kg
SK 3305.110 SK 3305.510	SK 3305.210 SK 3305.610	115 V, 50/60 Hz	11.0 A/ 12.5 A	42.0 A/ 46.0 A	20.0 A/ 20.0 A	100 %		880 W/1050 W 1040 W/1200 W	1500 W/1510 W 1230 W/1250 W	R134a, 600 g	25 bar	+20 – +55°C	< 64 dB (A)	IP 54/IP 34	400 x 950 x 260	46 kg
SK 3305.140 SK 3305.540	SK 3305.240 SK 3305.640	400 V, 3 – 50 Hz/ 460 V, 3 – 60 Hz	2.3 A/ 2.6 A	12.2 A/ 11.3 A	10.0 A*/ 10.0 A*	100 %		800 W/ 980 W 960 W/1150 W								42 kg
SK 3328.100 SK 3328.500	SK 3328.200 SK 3328.600	230 V, 50/60 Hz	6.9 A/ 8.5 A	22.0 A/ 26.0 A	16.0 A/ 16.0 A	100 %		900 W/1070 W 1130 W/1220 W								66 kg
SK 3328.110 SK 3328.510	SK 3328.210 SK 3328.610	115 V, 50/60 Hz	13.6 A/ 16.2 A	36.0 A/ 39.0 A	25.0 A/ 25.0 A	100 %		960 W/1130 W 1170 W/1290 W	2000 W/2350 W 1450 W/1690 W	R134a, 950 g	28 bar	+20 – +55°C	< 64 dB (A)	IP 54/IP 34	400 x 1580 x 290	73 kg
SK 3328.140 SK 3328.540	SK 3328.240 SK 3328.640	400 V, 3 – 50 Hz/ 460 V, 3 – 60 Hz	2.5 A/ 3.0 A	6.5 A/ 7.5 A	10.0 A*/ 10.0 A*	100 %		930 W/1150 W 1150 W/1400 W								67 kg
SK 3329.100 SK 3329.500	SK 3329.200 SK 3329.600	230 V, 50/60 Hz	8.0 A/ 10.0 A	21.0 A/ 21.0 A	16.0 A/ 16.0 A	100 %		1320 W/1550 W 1500 W/1880 W	2500 W/2750 W 1600 W/1750 W	R134a, 950 g	28 bar	+20 – +55°C	< 64 dB (A)	IP 54/IP 34	400 x 1580 x 290	69 kg
SK 3329.110 SK 3329.510	SK 3329.210 SK 3329.610	115 V, 60 Hz	16.0 A/ 21.0 A	44.0 A/ 42.0 A	25.0 A/ 25.0 A	100 %		1380 W/1600 W 1550 W/1940 W								76 kg
SK 3329.140 SK 3329.540	SK 3329.240 SK 3329.640	400 V, 3 – 50 Hz/ 460 V, 3 – 60 Hz	3.4 A/ 3.5 A	6.5 A/ 7.3 A	10.0 A*/ 10.0 A*	100 %		1300 W/1500 W 1550 W/1850 W	2500 W/2700 W 1900 W/1950 W							70 kg
SK 3332.140 SK 3332.540	SK 3332.240 SK 3332.640	400 V, 3 – 50 Hz/ 460 V, 3 – 60 Hz	3.8 A/ 3.9 A	9.2 A/ 11.0 A	10.0 A*/ 10.0 A*	100 %		1710 W/2110 W 1980 W/2450 W	4000 W/4400 W 3070 W/3570 W	R134a, 3000 g	28 bar	+20 – +55°C	< 65 dB (A)	IP 54/IP 34	500 x 1580 x 340	91 kg

Technische Änderungen vorbehalten. Technical modifications reserved. Sous réserve de modifications techniques. Technische wijzigingen voorbehouden.

Tekniska ändringar förbehålls. Rittal si riserva di apportare eventuali modifiche tecniche. Se reserva el derecho a realizar cambios técnicos.

* Motorschutzschalter, motor circuit breaker, disjoncteur-protecteur, motorbeveiligingsschakelaar, motorskyddsbytare, salvamotore, guardamotor.

Contents

- 1. Application
- 2. Technical data
- 3. Wall mounting
- 4. Safety notices
- 5. Electrical connection
- 6. Commencing operation and control behaviour
- 7. Supplementary functions
- 8. BUS system
- 9. Technical information
- 10. Handling Instructions
- 11. Scope of supply and guarantee
- 12. Display screen and system analysis
- 13. Comfort control programming

1. Application

Enclosure cooling units are designed and built to dissipate heat from enclosures by cooling the air inside the enclosure and protecting temperature-sensitive components. Enclosure cooling units are particularly suitable for a temperature range of +40°C to +55°C.

2. Technical data

(see table 2.1)

3. Wall mounting

According to production model, the wall-mounted unit can be mounted, installed, or partially installed, to choice (up to SK 3302.xxx and SK 3332.xxx). Cut out the sections and drill according to Fig. 3.1 at the fitting level.

3.1 Mounting

Cut the enclosed seals to the required length and attach to the unit in accordance with Fig. 3.2. Screw tapped pins into the blind nuts on the rear of the unit. The unit is then to be secured using washers and nuts.

3.2 Installation

First of all, the lamellar grid and belt are to be removed. The plug is to be removed from the display screen on the inner side of the belt. Cut the enclosed seals to the required length and attach to the unit in accordance with Fig. 3.2. Loosen the nuts and washers, then fasten the unit to the enclosure with the washers and nuts.

3.3 Partial installation

For partially built-in fitting, the housing must be divided (the unit cover is to be separated from the chassis. First of all, remove the lamellar grid and belt. This also requires the plug to be disconnected from the display screen on the inner side of the belt; loosen the nuts and washers. The cover can now be lifted, in order to disconnect the safety connection between cover and chassis, and separate the plug-in connection for the condenser fan. The unit cover can now be removed completely. It is now fitted to the enclosure (see Fig. 3.2) following the reverse sequence. The seals between chassis and cover are already available from production.

4. Safety notices

The following safety notices are to be observed in their entirety for the correct use of the equipment:

- To prevent the enclosure with the cooling unit fitted tipping over, it is essential that this be bolted to the floor.
 - A roller door is to be used to ensure problem-free opening and closing of the enclosure door.
 - A transportable enclosure with built-in cooling unit may only be produced if an additional transport anchorage to support the cooling unit is used.
- Prior to mounting, ensure that:
- the site for the enclosure, and hence the arrangement of the cooling unit, is selected so as to ensure good ventilation;
 - the location is free from excessive dirt and moisture;
 - the cutout for air extraction is located in the upper area of the enclosure;

- the mains connection ratings, as stated on the rating plate, are available;
- the ambient temperature does not exceed +55°C;
- the packaging shows no signs of damage. Traces of oil on damaged packaging are an indication of refrigerant loss and of leakage in the unit system. Any damage to the packaging may be the cause of subsequent malfunctions;
- the enclosure is sealed on all sides (IP 54). Condensation will occur if the enclosure is leaky;
- the separation of the units from one another and from the wall should not be less than 200 mm;
- air inlet and outlet are not obstructed on the inside of the enclosure;
- units are only fitted horizontally in the specified position. Max. deviation from the true horizontal: 2°;
- condensate drainage is provided (see 9.3);
- electrical connection and repair are carried out only by authorised personnel. Use only original replacement parts!
- to avoid an increase in condensation, a door-operated switch (e.g. PS 4127.000) should be used which will switch the cooling unit off when the enclosure door is opened (see 7.3);
- losses from the components installed in the enclosure must not exceed the specific refrigeration capacity of the cooling unit itself;
- the customer must not modify the cooling unit in any way.

5. Electrical connection

The connected voltage and frequency must correspond to the values stated on the rating plate. During commissioning, the data on the rating plate of the device shall apply. The cooling unit must be connected to the mains via an all-pin isolating device which ensures at least 3 mm contact opening when switched off. The unit must not have any additional temperature control connected up-stream at the supply end. The fans and compressors built into single and three-phase devices are intrinsically safe (thermal winding protection).

This also applies to SK 3304.110 / .510, SK 3305.110 / .510, SK 3328.110 / .510, SK 3329.110 / .510 transformer types, and to specially rated units which are also fitted with a transformer.

Line protection should be provided by means of the pre-fuse specified on the rating plate. Only one automatic cutout should be connected between the cooling unit and the power supply. A time-lag backup-fuse as specified on the rating plate (safety cutout K-characteristic or time-lag lead fuse) is mandatory.

The three phase supplies for the SK 3304. ... / SK 3305. ... / SK 3328. ... / SK 3329. ... / SK 3332. ... must be connected via a motor safety switch to a TN grid with grounded star point. Summation current as specified on the rating plate. Specially rated three-phase units must be protected by means of transformer safety switches (category AC-6A) specified on the rating plate.

For units designed for three phase 400/460 V, the rotary field or the absence of a phase is also monitored. If the rotary field is incorrect or a phase is absent, the unit will not run. Observe the relevant regulations during installation!

6. Commencing operation and control behaviour

Following the completion of mounting and a waiting period of approximately 30 minutes (to allow oil to collect in the compressor in order to ensure lubrication and cooling).

6.1 Basic control system

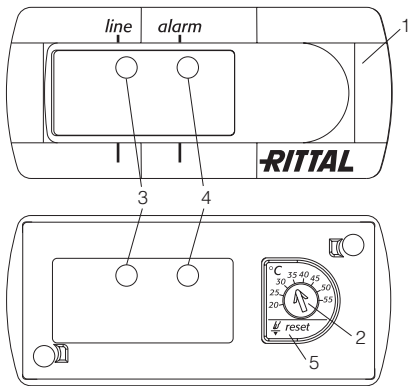
Version100 / .110 / .140

The cooling unit operates automatically, i.e. after electrical connection, the evaporator fan will run continuously to circulate the air inside the enclosure. The installed basic controller (setting the desired internal temperature; factory setting +35°C) provides automatic control switch-off of the cooling unit by the set value of the fixed switching difference of 5 K.

6.1.1 Temperature setting at the basic controller

First of all, the lamellar grid and belt with the incorporated display is to be removed from the unit. The display lock is then to be slackened and this is to be pushed forward off the lamellar grid/belt. Finally, the display screen (1) is carefully removed, e.g. with a screwdriver. The desired nominal temperature can be set on the potentiometer (2) which is now accessible. After the temperature has been set, the display of all screens in the lamellar grid is to be locked once more. The lamellar grid and belt are to be fastened to the unit again.

Fig. 6.1 Basic controller



6.1.2 Operation of the basic controller

The basic controller monitors and controls the cooling unit. After the supply voltage has been connected, the green LED (3) lights to indicate readiness for operation. The green LED is connected in series with a door limit switch, and flashes if the enclosure door is opened. The red LED (4) signals too high a temperature inside the enclosure; at 5 K above the set value, the red LED comes on permanently. In addition, the excess temperature message can still be queried via an integrated potential-free contact in the cooling unit connection terminal.

K1 fault signal relay with changeover contact

Terminal 3: NC (normally closed)

Terminal 4: C (connection between voltage

supply and fault signal relay)

Terminal 5: NO (normally open)

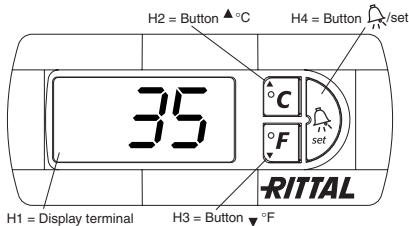
The definitions NC and NO refer to the de-energised state. As soon as voltage is applied to the cooling unit, the fault signal relay picks up. As a result, the relay contacts change their state (contact 3 – 4 open; contact 4 – 5 closed), representing the normal operating status of the cooling unit. As soon as a system message occurs or the voltage supply is interrupted, the relay drops out. (see also circuit diagrams on page 37).

The high pressure alarm in the cooling circuit is indicated by the red LED flashing. In this case, the unit is to be reset manually, by pressing the rubberized potentiometer display (5) for 3 seconds (see 6.1.1).

6.2 Comfort controller control

Version500 / .510 / .540

Fig. 6.2 Comfort controller



After electrical connection the internal fan turns on and circulates the enclosure air. This helps assure even temperature distribution within the enclosure. The condenser fan and compressor are controlled by the Comfort controller. The minimum break time is 3 min. The switching difference is 5 K, but can be set in the range 2 – 10 K. To avoid short switching cycles and hence the danger of inadequate or only partial cooling in some sections of the enclosure, the switching difference should be set to be only as low as necessary. For economic reasons (energy saving), the nominal value of the internal enclosure temperature T_i should also be set to be only as low as necessary.

6.2.1 Operation of the Comfort controller (see diagram 13.1 Programming)

The display terminal H1 consists of a 3 position 7-segment display which indicates the internal enclosure temperature in °C or °F (changeable) as well as any fault codes. The actual enclosure internal temperature is constantly displayed on H1. When a system message is generated, this alternates in the display with the current internal enclosure temperature. While programming the unit, the programming level and prescribed value are also displayed.

Test mode

By simultaneously holding down keys H2 (▲°C key) and H4 (↱/set key) for 5 seconds, the cooling unit will commence cooling operation, irrespective of the setpoint. The door limit switch function is disregarded in such cases. After approximately 5 minutes or upon reaching 15°C, the device deactivates cooling operation once again.

6.2.2 Programming (see 13.1 Comfort controller programming)

In the EEPROM of the Comfort controller various parameters are stored which can be changed by using the buttons H2, H3 and H4. 24 changeable parameters can be set via 24 programme levels in the stated ranges (max. and min. values). To this end, input code "22" is required (see diagram 13.1). The H2, H3 and H4 keys are multiple occupancy. To access programming mode, set 5 sec is to be pressed and held down. Primary occupancy (only diode displays): H2: °C, H3: °F, H4: Alarm Secondary occupancy (for programming): H2: Arrow upwards, H3: Arrow downwards, H4: set Programming via diagnostic software (program no. SK 3159.100) is also an option. The programming cable interface is the connecting cable plug, located on the rear of the Comfort controller. To do this, the lamellar grid and the belt also are to be removed if necessary.

6.2.3 System messaging equipment

All system messages at the cooling unit are recorded and displayed as an error number by H1. System messages alternate in the display with the current internal enclosure temperature. (see 12. Comfort control fault display and error analysis).

6.2.3.1 System message contacts (K1 and K2; potential free)

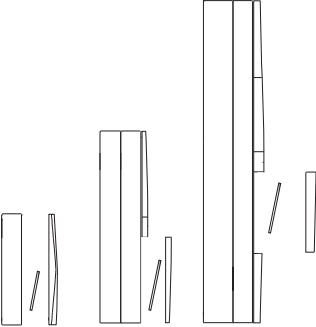
Both relays are normally closed. All system messages assigned to an individual relay result in it opening. If the control voltage fails, this also results in the relay opening and can hence be captured and documented in the log file. According to individual weighting, all system messages can be assigned to both relays or suppressed. Terminal strip X3 provides the connection. See the wiring diagram for contact data and occupancy. K1/K2 fault signal relay (normally open contact): Terminal 3: Fault signal relay 2 Terminal 4: Voltage supply connection for both fault signal relays Terminal 5: Fault signal relay 1 (see circuit diagrams on page 37).

7. Supplementary functions

7.1 Filter mat use and changing

The PU foam filter mat available as an accessory is coarse and filters large dust particles or fluff from the air. Metallic filter mats are used to trap oil condensate. These are also available as an accessory. Subject to the suction of the blower being high enough, fine dust is blown through the filter mat and the external circuit of the unit. This does not affect the unit's operation.

Fig.7.1 Filter mat replacement



7.2 Filter mat monitoring

Function of the filter mat monitor: The filter mat is monitored for soiling by measuring the temperature difference in the external circulation of the cooling unit (see Diagram 13.1: Programming on page 45). In the event of any filter mat soiling, the temperature difference will increase. The nominal value of the temperature difference in the external circuit is matched to the relevant operating points in the characteristic fields. Hence there is no requirement to adjust the nominal value for different unit operating points.

7.3 Door limit switch S1 (supplied by customer)

Where a door limit switch is used and the enclosure door is open (contact is closed when door is open), the cooling unit (fans and condenser) will switch off after approx. 15 s. This only applies to devices with a comfort controller and devices with a three-phase connection to the basic controller. For devices with a basic controller (115 V, 230 V and 400 V, 2~), the internal fan is not switched off when using a door limit switch. Thereby avoiding an increase in condensation while the door is open. To avoid cyclic operation, switch-on of condenser and external fan is delayed by about 3 minutes after the cooling unit has been switched off. The internal fan will start up after about 15 s on closure of the door. Connection is made at the terminals 1 and 2. The extra low voltage is supplied by the internal power pack, current is approx. 30 mA DC. Each door limit switch must only be assigned to one cooling unit. Several door limit switches may be operated on one cooling unit (parallel connection). The minimum cross-section of the connection cable is 0.3 mm² for a cable length of 2 m. The resistance of the door limit switch contact must not exceed a maximum of 50 Ω. Connect the door limit switch free from potential only, no external voltage!

7.4 Interface X3 (option) (Connector X3)

Note!

The electrical signals at the interface are of an extra-low voltage (not extra-low safety voltages to EN 60 335). The 9-pin SUB-D connector X3 can be used to connect additional interface cards for integrating cooling units with higher level monitoring systems. These cards are also available as an accessory. (Model No.: Interface card SK 3124.200).

8. BUS system

(Model No.: Master-slave cable SK 3124.100)

8.1 General

The BUS system allows a maximum of 10 cooling units to be interconnected. As a result, the following functions are available to the operator: ● Parallel unit control (the cooling units in the network can be simultaneously switched on and off) ● Parallel door status messages ("door open") ● Parallel collective fault message

The data exchange is carried out using master-slave cables (shielded, two-wire leads). All units are assigned an address. This address also includes the ID for "master" or "slave".

8.2 Installation notices for the X2 interface (Jack X2)

Note!

The electrical signals at the interface are of an extra-low voltage (not extra-low safety voltages to EN 60 335). Always heed the following notes! ● De-energise the cooling units to be connected. ● Ensure proper electrical insulation. ● Make sure the cables are not laid in parallel to power lines. ● Make sure that the lines are short.

8.3 Programming the cooling unit

See diagram 13.1 for details on programming.

ID's:

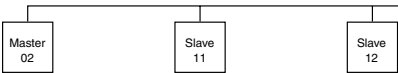
Master cooling unit		Slave cooling unit	
00	Basic state	00	Basic state
01	Master with 1 slave	11	Slave with address 1
02	Master with 2 slaves	12	Slave with address 2
03	Master with 3 slaves	13	Slave with address 3
04	Master with 4 slaves	14	Slave with address 4
05	Master with 5 slaves	15	Slave with address 5
06	Master with 6 slaves	16	Slave with address 6
07	Master with 7 slaves	17	Slave with address 7
08	Master with 8 slaves	18	Slave with address 8
09	Master with 9 slaves	19	Slave with address 9

Note

Only one unit may be configured as master; the address ID must match the number of slave units. The individual slave units must have different addresses; the addresses must be in ascending order (without gaps in between).

Example:

1 master cooling unit with 2 slave cooling units



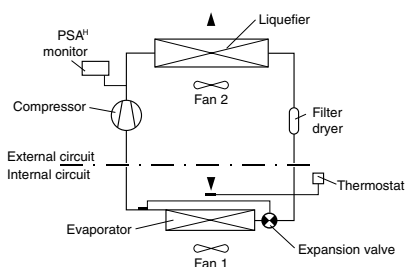
(see wiring example of master/slave and door limit function, page 42/43).

9. Technical information

The cooling unit (compressor refrigeration unit) consists of four main components: the coolant compressor, evaporator, condenser, and the control or expansion valve, which are connected by suitable pipework. This circuit is filled with a readily boiling substance, the refrigerant. The R134a (CH_2FCF_3) refrigerant is free from chlorine. It has an ozone destroying potential (ODP) of 0 and is therefore environmentally friendly. A filter dryer which is integrated in the hermetically sealed cooling circuit, provides effective protection against moisture, acid, dirt particles, and foreign bodies within the cooling circuit.

9.1 Operation of the cooling unit

Fig. 9.1 Operation of the cooling unit



The compressor takes the gaseous refrigerant from the evaporator and compresses it to a higher pressure in the condenser. During this process the temperature of the refrigerant rises above the ambient temperature and heat can be dissipated to the environment via the surface of the condenser. Then the refrigerant is liquefied and, by means of a thermostatically controlled expansion valve, returned to the evaporator, where it evaporates at low pressure. The heat required for complete evaporation is drawn from the enclosure interior causing it to cool down. The cooling cycle is thus completed, the aforementioned process of the heat transfer starts afresh.

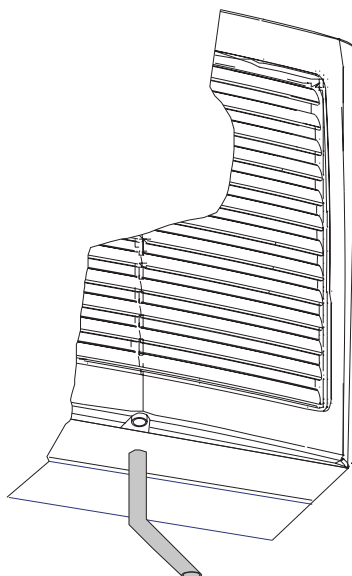
9.2 Safety equipment

The cooling circuit of the cooling unit has a component-tested, high-pressure monitor to EN 12 263, which is set to maximum PS (allowed pressure) and operates via an automatic reset device in case of recurring pressure drop. Temperature monitoring will prevent the evaporator from icing up. If there is a risk of icing up, the compressor is switched off and automatically switched on again at higher temperatures. The refrigerant compressor and the fans are equipped with thermal winding protection switches against excess current and excess temperatures.

9.3 Condensate discharge

A drain pipe fitted to the evaporator divider panel ensures that any condensate which may form on the evaporator (at high air humidity, low temperatures inside the enclosure) is drained away from the bottom of the unit. For this purpose, a length of hose should be fitted to the condensate pipe connection piece (see fig. 9.3, if appropriate, remove the grille). The condensate must be able to run off freely. If the condensate is to be drained off over a greater distance, then care must be taken that the hose is free from kinks and a check for correct drainage made.

Fig. 9.3 Condensate discharge



10. Handling Instructions

Storage, maintenance, transport and disposal

As a maintenance-free, hermetically sealed system, the cooling circuit has been filled in the factory with the required amount of refrigerant, and tested for leaks and subjected to a function trial run.

The installed maintenance-free fans use ball bearings, they are protected against moisture and dust, and are fitted with a temperature monitor. The life expectancy is at least 30,000 operating hours. The cooling unit is thus largely maintenance free. All that may be required from time to time is that the components of the external air circuit are cleaned by compressed air.

The use of a filter mat is recommended only if large particles of lint are present in the air, so that blockage of the condenser is prevented. (Filter mat replacement, fig. 7.1). Note: Prior to any maintenance work, the power to the cooling unit must be disconnected.

Storage temperature: The cooling units must not be exposed to temperatures above +70°C.

Transport position: The cooling units must always be transported upright.

Waste disposal: The closed cooling circuit contains refrigerant and oil which must be correctly disposed of for the protection of the environment. Disposal can be carried out at Rittal.

We reserve the right to make technical modifications.

11. Scope of supply and guarantee

- 1 cooling unit, ready for connection
- 1 drilling template
- 1 sealing tape
- 1 set of mounting and operating instructions
- 1 plug-in terminal strip

Guarantee:

This unit is covered by a 1-year guarantee from the date of supply, subject to correct usage (see also Safety notices under heading 4.). Within this period, the returned unit will be repaired in the factory or replaced free of charge. The cooling unit is to be used for the cooling of enclosures only. If it is connected or handled improperly the manufacturer's guarantee does not apply and in this case we are not liable for any damage caused.

12. Comfort control display screen and system analysis

Alarm no.	System message	Cause	Remedy
A01	Open the enclosure door	Door open or door limit switch not correctly positioned	Close door, position door limit switch correctly, check connection if necessary
A02	Internal temperature of enclosure too high	Cooling capacity too low/unit undersize Sequence errors in messages A03 to A17	Check cooling capacity
A03	Filter monitoring	Filter mat soiled	Clean or exchange
A04	Ambient temperature too high/too low	Ambient temperature outside the permissible operating range (+10°C to +60°C)	Raise or lower the ambient temperature (e.g. heat or ventilate the room)
A05	Icing hazard	Icing hazard display according to operating mode	Increase the nominal value of the internal enclosure temperature
A06	PSA ^H monitor	Ambient temperature too high	Unit operating limits exceeded
		Condenser soiled	Clean
		Filter mat soiled	Clean or exchange
		Condenser fan defective	Exchange
		E-valve defective	Repair by refrigeration engineer
		PSA ^H monitor fault	Refrigeration engineer to exchange
A07	Evaporator	Refrigerant fault	Repair by refrigeration engineer
A08	Condensate warning	Only in units with optional condensate evaporation	Check condensate drainage; repair kinks or blockages in the hose as necessary
A09	Condenser fan	Blocked or defective	Clear the blockage; exchange if necessary
A10	Condenser fan	Blocked or defective	Clear the blockage; exchange if necessary
A11	Compressor	Compressor overloaded (internal winding protection)	Unit switches in again independently
		Defective (check by measuring the winding resistance)	Refrigeration engineer to exchange
A12	Condenser temperature sensor	Open or short circuit	Exchange
A13	Ambient temperature sensor	Open or short circuit	Exchange
A14	Icing temperature sensor	Open or short circuit	Exchange
A15	Condensate warning temperature	Open or short circuit	Exchange
A16	Internal temperature sensor	Open or short circuit	Exchange
A17	Phase monitoring	Incorrect rotary field/phase absent	Exchange two phases
A18	EPROM error		
A19	LAN/Master-Slave	Master and slave not linked	Setting or check cable
A20	Voltage drop	Fault display is not illustrated	Event is saved in the log file
E0	Display signal	Connection problem (display and control board)	Reset by means of power supply (OFF/ON)
		Cable faulty; plug connection loose	Replacement of the boards

The fault displays A03, A06 and A07 are to be reset manually. To do this, hold down the ▲ and ▼ keys simultaneously for 5 seconds.

13. Comfort control programming

Level	Display screen	changeable parameter	min. value	max. value	Factory setting	Explanation
1	St	Nominal value of the internal enclosure	20	55	35	The nominal value of the internal enclosure temperature is set at the factory to 35°C and is variable over a 20°C range to 55°C.
2	Fi	Filter mat monitoring	10	60 (99 = off)	99	For activation of filter mat monitoring, the display is to be set to a minimum of 10 K above the temperature difference shown in programming mode "Fi"; filter mat monitoring is switched off at the factory. (Display 99 = off).
3	Ad	Master-slave programming	0	19	0	see 8.3 "Cooling unit programming"
4	CF	Temperature conversion	0	1	0	The temperature display can be switched from °C (0) to °F (1). The corresponding LED displays the current unit of temperature.
5	H1	Temperature switching	2	10	5	The cooling unit is set in the factory for a switching cycle of 5 K. This parameter should only be changed after agreement with the manufacturer.
6	H2	Difference for error message 2	3	15	5	If the internal enclosure temperature exceeds the set value by more than 5 K, then error message 2 (internal enclosure temperature too high) appears on the display terminal. The difference of 5 K can be changed if required over the range 3 K to 15 K.
7	A1	Relay connection	0	2	0	The fault messages (1 to 19) depicted in the display can also be evaluated using two potential-free fault message relays. Comfort control provides the option to group fault messages accordingly and assign them to both fault message relays. The value "1" is to be programmed for assignment and evaluation of the relevant fault message via relay 1. If the fault is to be evaluated via relay 2, then the value "2" is to be programmed accordingly. If the value is set to "0", the system message appears only in the display. This accordingly enables the user to define the system weighting himself, by apportioning between relays 1 and 2.
8	A2	Relay connection	0	2	0	
9	A3	Relay connection	0	2	0	
10	A4	Relay connection	0	2	0	
11	A5	Relay connection	0	2	0	
12	A6	Relay connection	0	2	1	
13	A7	Relay connection	0	2	2	
14	A8	Relay connection	0	2	1	
15	A9	Relay connection	0	2	1	
16	A10	Relay connection	0	2	1	
17	A11	Relay connection	0	2	2	
18	A12	Relay connection	0	2	1	
19	A13	Relay connection	0	2	1	
20	A14	Relay connection	0	2	1	
21	A15	Relay connection	0	2	1	
22	A16	Relay connection	0	2	1	
23	A17	Relay connection	0	2	1	
24	A18	Relay connection	0	2	0	
25	A19	Relay connection	0	2	0	

D Anschlussschema Comfortcontroller

A1	= Leistungsplatine
A2	= Anzeigeterminal
A3	= Anlassrelais und R-C Glied
B1	= Temperaturfühler Innentemperatur
B2	= Temperaturfühler Vereisungsschutz
B3	= Temperaturfühler außen 1
B4	= Temperaturfühler außen 2
B5	= Kondensatwarnsensor
C1-C4	= Betriebskondensatoren
F2	= PSA ^H -Wächter, (bei 3302.1x0 kein Pressostat sondern Brücke)
F3	= Thermokontakt Verdichter
K1	= Relais Sammelstörung 1
K2	= Relais Sammelstörung 2
L1	= LED Betrieb grün
L2	= LED Alarm rot
M1	= Verdichter
M2	= Verflüssigerventilator
M4	= Verdampferventilator
R1	= Potentiometer zur Setpoint-Einstellung
S1	= Türeenschalter (ohne Türeenschalter Klemme 1, 2 offen)
T1	= Trafo
X1	= Hauptanschlussklemmleiste
X2	= Master/Slave-Anschluss
X3	= Optionale Schnittstelle

GB Wiring diagram comfort controller

A1	= Power PCB
A2	= Display terminal
A3	= Start-up relay and R-C component
B1	= Internal temperature sensor
B2	= Icing hazard temperature sensor
B3	= Temperature sensor, external 1
B4	= Temperature sensor, external 2
B5	= Condensate warning sensor
C1-C4	= Operating capacitors
F2	= PSA ^H monitor, (in the 3302.1x0 no pressure-sensitive switch but a bridge)
F3	= Compressor thermal contact
K1	= Relay collective fault 1
K2	= Relay collective fault 2
L1	= Green operating LED
L2	= Red alarm LED
M1	= Compressor
M2	= Condenser fan
M4	= Evaporator fan
R1	= Setpoint adjustment potentiometer
S1	= Door limit switch (without door operated switch terminal 1, 2 open)
T1	= Transformer
X1	= Main terminal strip
X2	= Master/slave connection
X3	= Optional interface

F Schéma électrique régulateur confort

A1	= Platine de puissance
A2	= Display terminal
A3	= Relais de démarrage temporisé
B1	= Sonde de température intérieure
B2	= Sonde de veille anti-givre
B3	= Sonde de température extérieure 1
B4	= Sonde de température extérieure 2
B5	= Avertisseur de condensats
C1-C4	= Condensateurs de régime
F2	= Témoin pression PSA ^H , (sauf 3302.1x0 : le pressostat est remplacé par un pont)
F3	= Contact bilame du compresseur
K1	= Relais défauts 1
K2	= Relais défauts 2
L1	= LED fonction – vert
L2	= LED alerte – rouge
M1	= Compresseur
M2	= Ventilateur du condenseur
M4	= Ventilateur de l'évaporateur
R1	= Potentiomètre pour l'étalement
S1	= Interrupteur de porte (en absence d'interrupteur bornes 1, 2 ouvertes)
T1	= Transformateur
X1	= Bornier principal
X2	= Port maître/esclave
X3	= Interface (option)

NL Aansluitschema comfortcontroller

A1	= Hoofdstroomprint
A2	= Display
A3	= Startrelais en R-C-element
B1	= Temperatuursensor interne temp.
B2	= Temperatuursensor ijsvorming
B3	= Temperatuursensor buiten 1
B4	= Temperatuursensor buiten 2
B5	= Condens waarschuwingssensor
C1-C4	= Motorcondensator
F2	= PSA ^H -pressostaat, (bij 3302.1x0 geen pressostaat maar brug)
F3	= Thermisch relais compressor
K1	= Relais verzamelstoring 1
K2	= Relais verzamelstoring 2
L1	= LED bedrijf, groen
L2	= LED alarm rood
M1	= Compressor
M2	= Condensorventilator
M4	= Verdamperventilator
R1	= Potmeter voor instellen gewenste waarde
S1	= Deurschakelaar (zonder deurschakelaar klem 1, 2 open)
T1	= Transformator
X1	= Hoofdaansluitklemmenstrook
X2	= Master/slave-aansluiting
X3	= Optionele interface

S Anslutningsschema comfortcontroller

A1	= Drivkort
A2	= Display terminal
A3	= Startrelä och R-C länk
B1	= Temperaturgivare innetemperatur
B2	= Temperaturgivare nedisningsrisk
B3	= Temperaturgivare yttre 1
B4	= Temperaturgivare yttre 2
B5	= Kondensvarningssensor
C1-C4	= Startkondensator
F2	= PSA ^H -vakt, (vid 3302.1x0 ingen pressostat utan bygling)
F3	= Termokontakt kompressor
K1	= Samlingsrelä felsignaler 1
K2	= Samlingsrelä felsignaler 2
L1	= LED drift grönt
L2	= LED alarm rött
M1	= Kompressor
M2	= Kondensorfläkt
M4	= Förångarfläkt
R1	= Potentiometer för setpoint-inställning
S1	= Dörrströmbrytare (utan dörrströmbrytarklamma 1, 2 öppna)
T1	= Trafo
X1	= Huvudkontaktplint
X2	= Master/Slave-anlutning
X3	= Optional anslutning

I Schema allacciamenti controllore Comfort

A1	= Scheda di potenza
A2	= Display terminale
A3	= Relais di avviamento
B1	= Sonda temperatura interna
B2	= Sonda temperatura, pericolo di formazione di ghiaccio
B3	= Sonda temperatura esterna 1
B4	= Sonda temperatura esterna 2
B5	= Sensore allarme condensa
C1-C4	= Condensatore d'esercizio
F2	= Controllo PSA ^H , (eccetto 3302.1x0)
F3	= Contatto termico compressore
K1	= Relè segnalatore guasti 1
K2	= Relè segnalatore guasti 2
L1	= LED esercizio verde
L2	= LED allarme rosso
M1	= Compressore
M2	= Ventilatore del condensatore
M4	= Ventilatore dell'evaporatore
R1	= Potenzziometro per setpoint
S1	= Interruttore della portina (senza interruttore i morsetti 1, 2 sono aperti)
T1	= Trasformatore
X1	= Morsettiera a raccordo principale
X2	= Raccordo Master/Slave
X3	= Interfaccia opzione

E Esquema de conexiones del controlador Confort

A1	= Pletina de potencia
A2	= Pantalla indicadora
A3	= Relé de arranque y fusible R-C
B1	= Sonda térmica de la temp. en el interior del armario
B2	= Sonda térmica protección contra congelación
B3	= Sonda térmica exterior 1
B4	= Sonda térmica exterior 2
B5	= Sensor de condensación
C1-C4	= Condensador electrolítico de servicio
F2	= Presostato PSA ^H , (en 3302.1x0 sin presostato sino puente)
F3	= Contacto térmico compresor
K1	= Relé de fallo 1
K2	= Relé de fallo 2
L1	= LED servicio verde
L2	= LED alarma rojo
M1	= Compresor
M2	= Ventilador del condensador
M4	= Ventilador del evaporador
R1	= Potenciometro para ajuste de valores
S1	= Interruptor de puerta (sin interruptor final borne 1, 2 abierto)
T1	= Transformador
X1	= Regleta de bornes principal
X2	= Conexión maestro/esclavo
X3	= Interfaz opcional

J 配線図

A1	= 電源基板
A2	= ディスプレイターミナル
A3	= 起動リレー と RC
B1	= 温度センサー 内部温度
B2	= 温度センサー 凍結防止
B3	= 温度センサー 外部温度 1
B4	= 温度センサー 外部温度 2
B5	= 凝縮水センサー
C1-C4	= コンデンサー
F2	= 圧カスイッチ (3302.1x0 は圧カスイッチなし。ブリッジのみ)
F3	= サーマルスイッチ 圧縮機
K1	= リレー 異常時外部出力 1
K2	= リレー 異常時外部出力 2
L1	= LED 運転 (緑)
L2	= LED アラーム (赤)
M1	= 圧縮機
M2	= 凝縮器ファン 1
M3	= 凝縮器ファン 2
M4	= 蒸発器ファン
R1	= 設定温度調整用つまみ
S1	= ドアスイッチ (非接続時、端子 1 と 2 はオープン)
T1	= トランス
X1	= 主端子台
X2	= マスター/スレーブ接続
X3	= オプションインターフェイス

Anschlussschema

Wiring diagram

Schéma électrique

Aansluitschema

Anslutningsschema

Schema allacciamenti

Esquema de conexiones

配線図

Kontaktdaten K1

Contact Data K1

Caracteristiques des contacts K1

Kontaktgegevens K1

Kontaktdata K1

Caratteristiche dei contatti K1

Características del contacto K1

K1の接点データ

AC cos ϕ = 1	DC L/R = 40 ms
I max. = 5 A U max. = 230 V	I min. = 10 mA U max. = 100 V → I max. = 200 mA U max. = 20 V → I max. = 5 A

Technische Daten siehe Typenschild

Technical data see name plate

Données techniques voir plaque signalétique

Technische gegevens zie typenplaatje

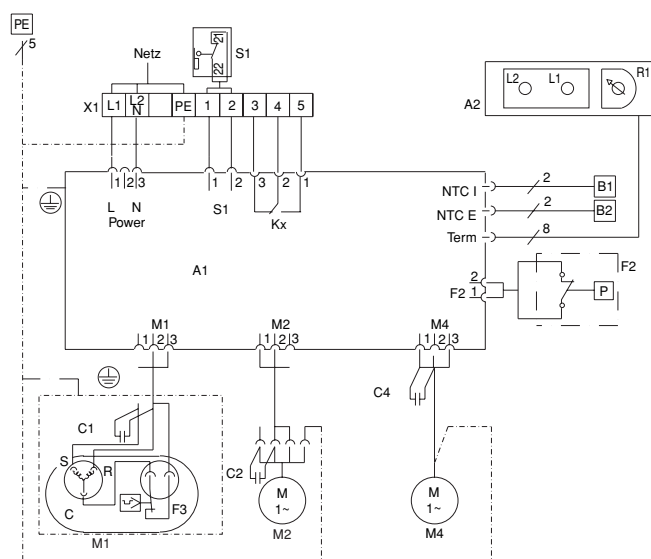
Tekniska data se typskytt

Caratteristiche tecniche vedi targhetta

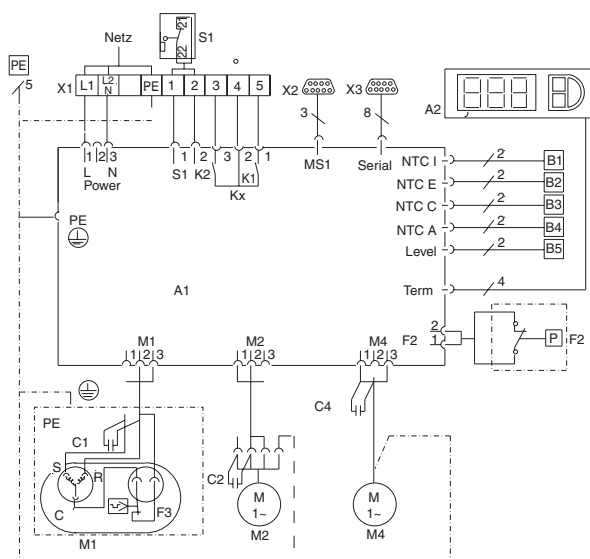
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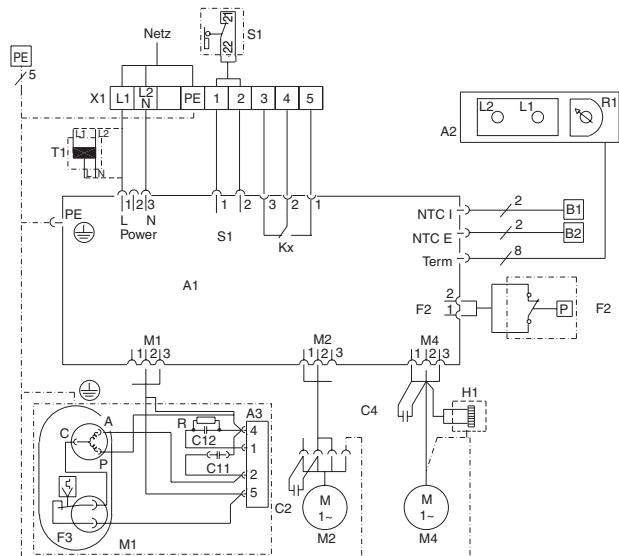
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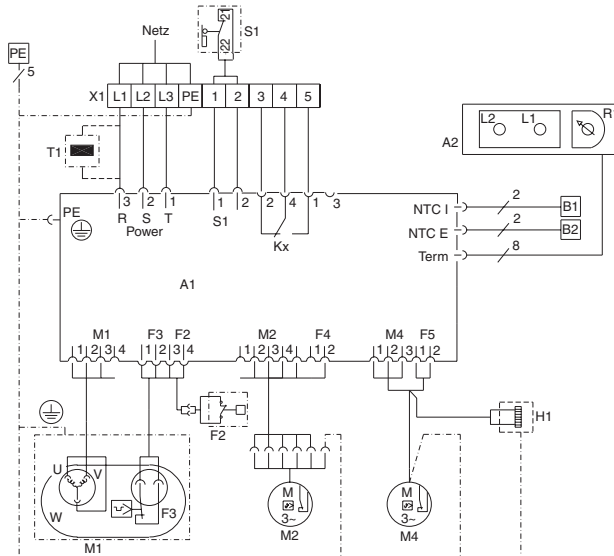
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SK 3303.600 / .610



**SK 3304.100 / .110, SK 3305.100 / .110,
SK 3328.100 / .110, SK 3329.100 / .110,
SK 3304.200 / .210, SK 3305.200 / .210,
SK 3328.200 / .210, SK 3329.200 / .210**



**SK 3304.140, SK 3305.140,
SK 3328.140, SK 3329.140,
SK 3304.240, SK 3305.240,
SK 3328.240, SK 3329.240**



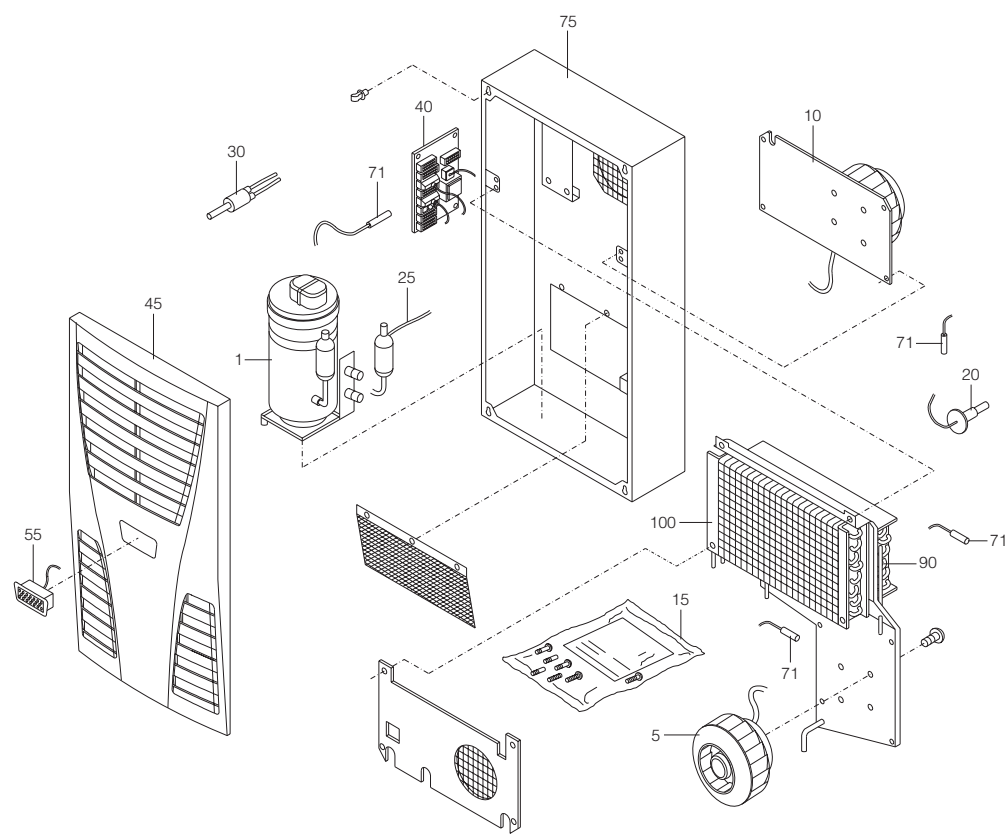
Position Item Pos. Pos. Pos. Posición	Ersatzteil- liste	Spares list	Liste de pièces détachées	Lijst reserve- delen	Reserv- delslista	Lista dei pezzi di ricambio	Lista de piezas de repuesto	スベアパーツ 一覧表
	D	GB	F	NL	S	I	E	J
	Bezeichnung	Description	Signification	Benaming	Benämning	Descrizione	Descripción	名称
1	Verdichter	Compressor	Compresseur	Kompressor	Kompressor	Compressore	Compresor	圧縮機（コンプレッサー）
5	Verflüssiger- ventilator	Condensing fan	Ventilateur du condenseur	Condensor- ventilator	Kondensorfläkt	Ventilatore del condensatore	Ventilador del condensador	凝縮器ファン
10	Verdampfer- ventilator	Evaporator fan	Ventilateur de l'évaporateur	Verdampfer- ventilator	Förångarfläkt	Ventilatore dell'evaporatore	Ventilador del evaporador	蒸発器ファン
15	Versandbeutel	Dispatch bag	Pochette d'accessoires	Zakje toebehoren	Tillbehörspåse	Sacchetto accessori	Bolsa de accesorios	標準アクセサリキット
20	Expansionsventil	Expansion valve	Soupape de détente	Expansieventiel	Expansionsventil	Valvola d'espansione	Válvula de expansión	膨張弁
25	Filtertrockner	Filter dryer	Assècheur de filtre	Filterdroger	Filtertork	Filtro essicatore	Secador del filtro	フィルタドライヤー
30	PSA ^H -Wächter	PSA ^H monitor	Témoin pression PSA ^H	PSA ^H -pressostaat	PSA ^H -vakt	Controllo PSA ^H	Presostato PSA ^H	圧カスイッチ
40	Controller-Box	Controller box	Processeur box	Controller E-box	Kontroll box	Box controller	Carcasa controlador	制御基板
45	Lamellengitter 1	Louvred grille 1	Grille à lamelles 1	Rooster 1	Lamellgaller 1	Griglia a lamelle 1	Rejilla 1	プラスチックルーバー1
46	Lamellengitter 2	Louvred grille 2	Grille à lamelles 2	Rooster 2	Lamellgaller 2	Griglia a lamelle 2	Rejilla 2	プラスチックルーバー2
50	Abdeckblende	Infill panel	Plastron	Afdekplaat	Täckplåt	Copertura cieca	Pantalla cubierta	センターカバー
55	Anzeige	Display	Affichage	Display	Display	Comando	Indicación	ディスプレイ
65	Folientastatur	Membrane keyboard	Clavier membrane	Folietoetsenbord	Folietangenter tryckknappar	Tastiera a membrana	Teclado de membrana	タッチパネル
66	Abdeckfolie	Covering membrane	Feuille de recouvrement	Afdeckfolie	Täckfolie	Lamina di copertura	Lámina cubierta	タッチパネル用フイルムカバー
71	Temperaturfühler	Temperature sensor	Sonde de température	Temperatuur- sensor	Temperaturgivare	Sonda di temperatura	Sonda térmica	温度センサ
75	Haube	Cover	Couvercle	Afdekkap	Huv	Calotta	Capucha	カバー
80	Transformator	Transformer	Transformateur	Trafo	Trafo	Trasformatore	Transformador	トランス
90	Verdampfer	Evaporator	Évaporateur	Verdamper	Förångare	Evaporatore	Evaporador	蒸発器（エバポレーター）
100	Verflüssiger	Condenser	Condenseur	Condensor	Kondensor	Condensatore	Condensador	凝縮器

Bei Bestellung unbedingt angeben
Typ:
Fabrikations-Nr.:
Herstelldatum:
Ersatzteil-Nr.:

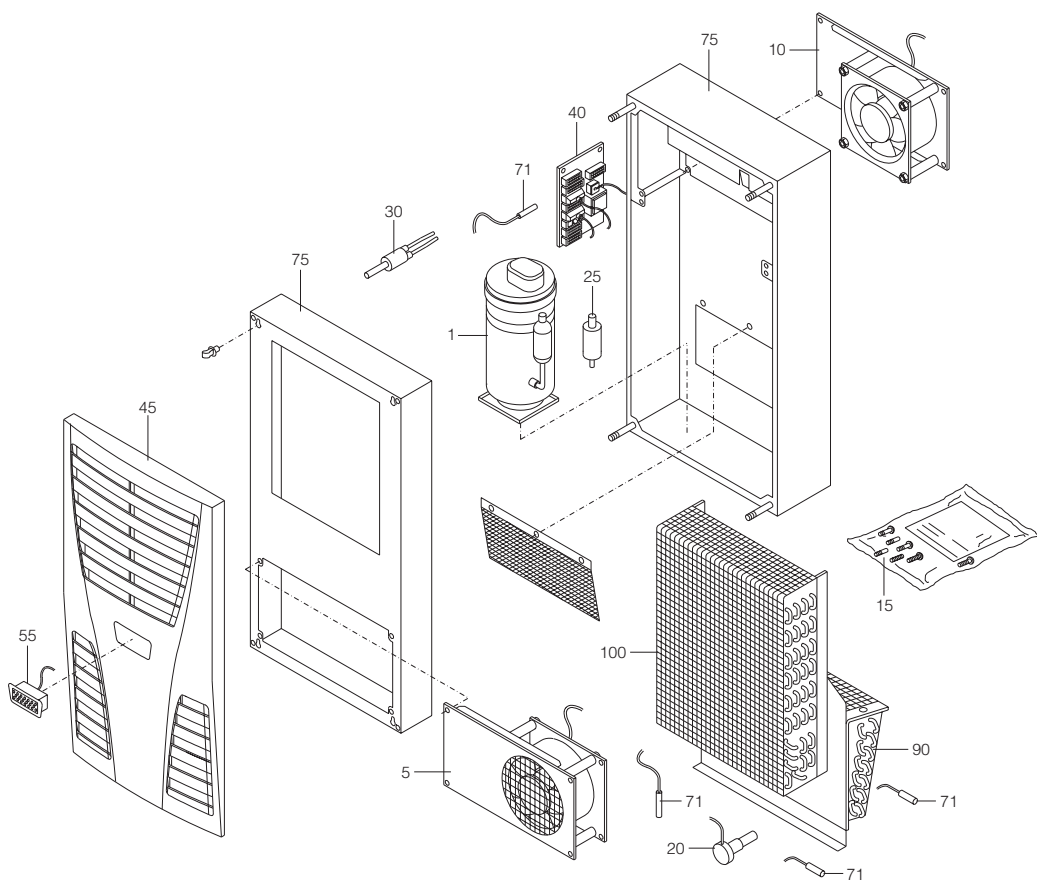
Absolutely necessary in case of order
Type:
Fabrication no.:
Manufacturing date:
Spare part no.:

スベアパーツのご発注時に、必ずご記入ください。
製品番号： SK
製造番号：
製造日時：
スベアパーツ番号：

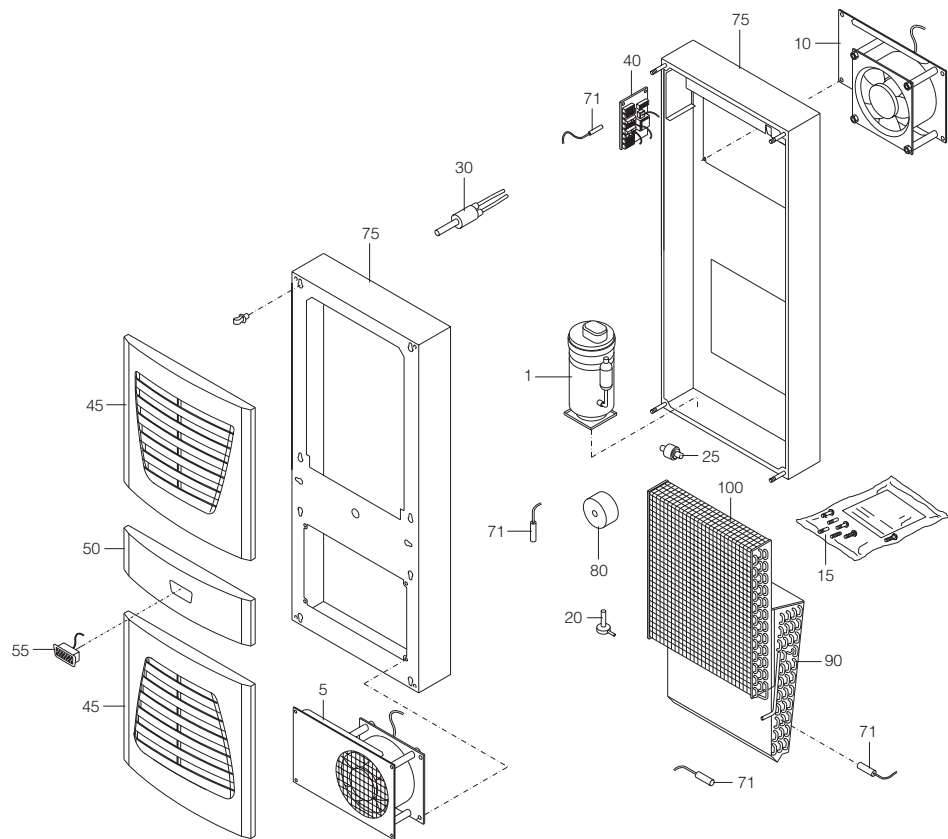
SK 3302.xxx



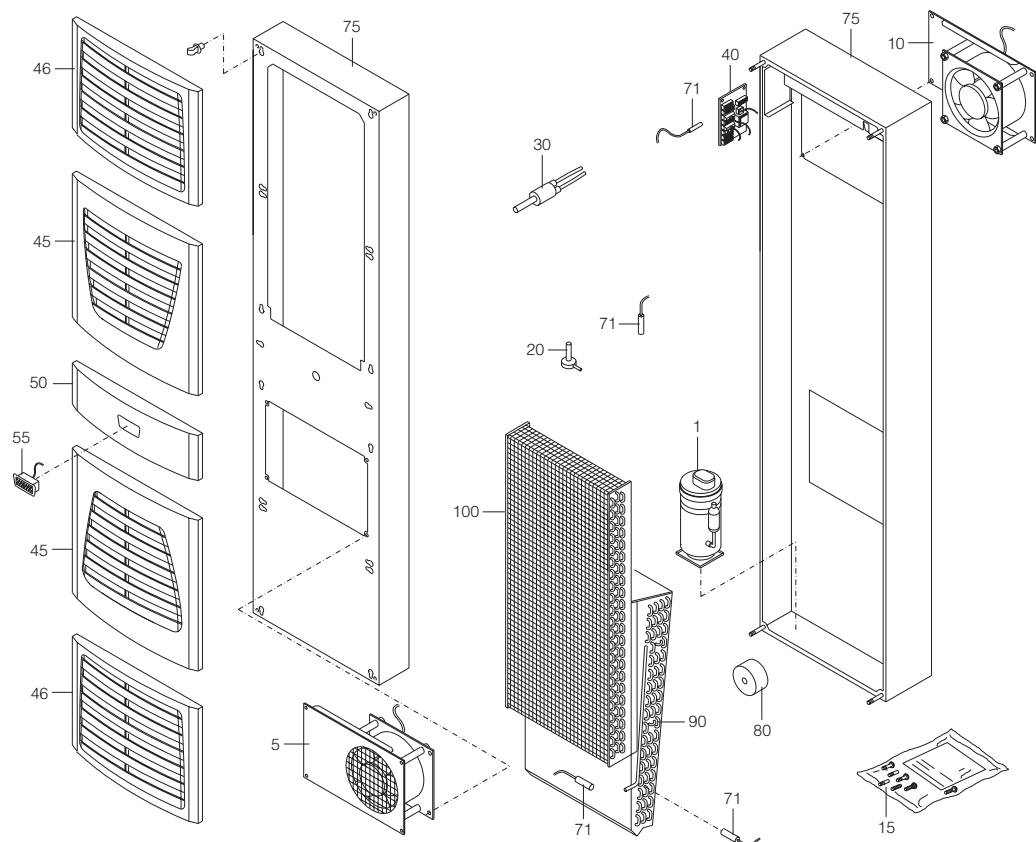
SK 3303.xxx



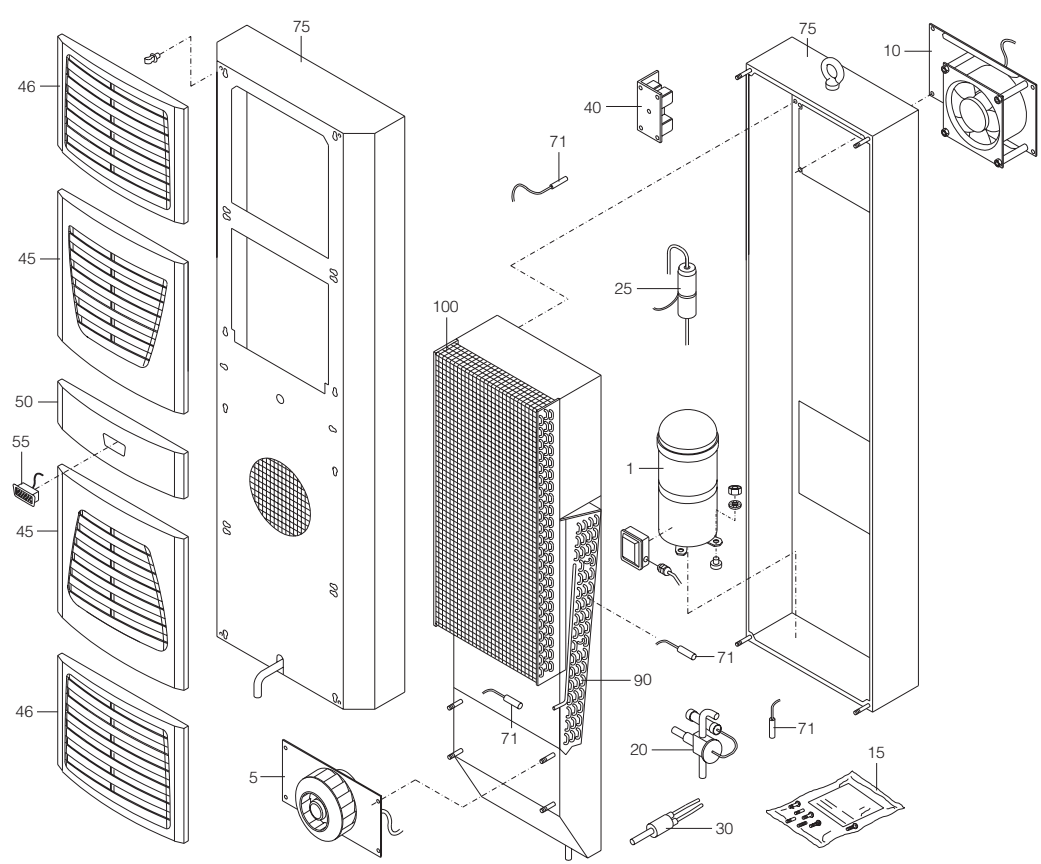
SK 3304.xxx
SK 3305.xxx



SK 3328.xxx
SK 3329.xxx



SK 3332.xxx



Anschlussbeispiel: Master-Slave-Betrieb

Connection example: Master/slave operation

Schéma d'installation typique : régime maître-esclave

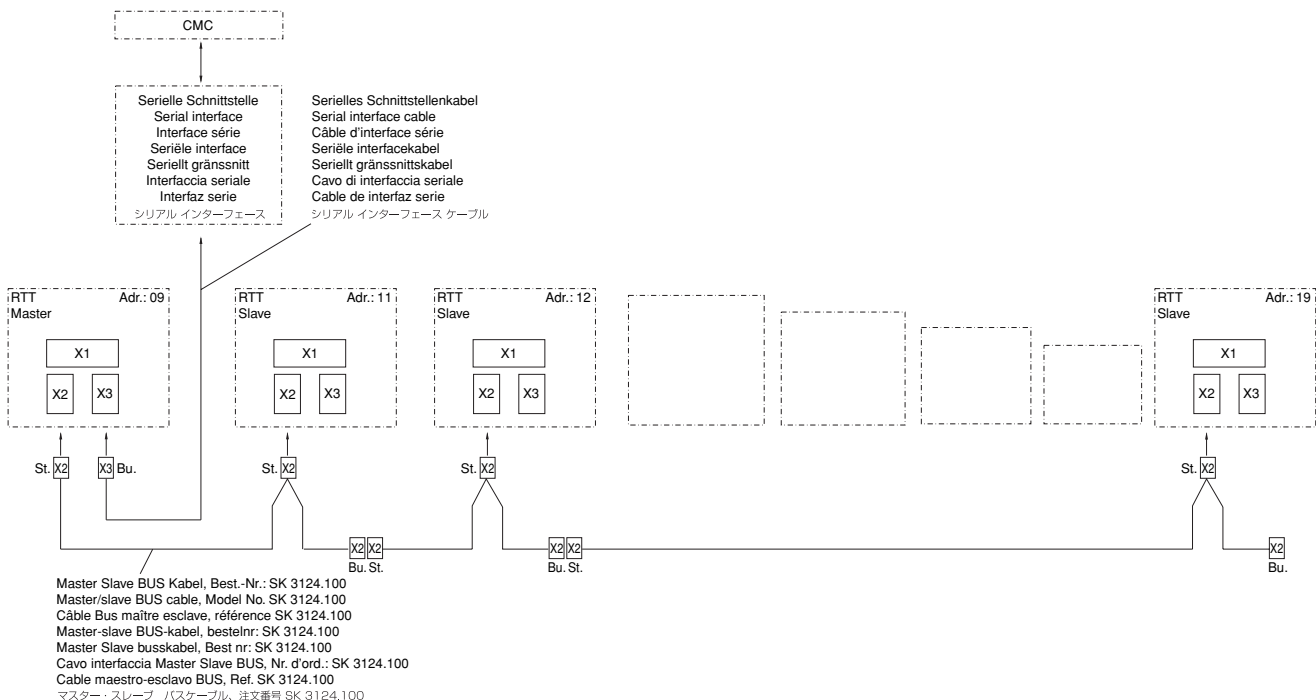
Aansluitvoorbeeld: master-slave functie

Uppkopplingsexempel: Master-Slave installation

Esempi di connessione: funzione Master-Slave

Ejemplo de conexión: Funcionamiento maestro-esclavo

マスター・スレーブ運転の接続例



D Beschreibung
Die Adresse de

Beschreibung

Die Adresse des Masters ist abhängig von der Anzahl der angeschlossenen Slave-Geräte (09 = Master mit 1 Slave-Gerät). Die Adresse eines Slave-Gerätes beginnt immer mit der 1. Die 2. Zahl stellt die eigentliche Adressierung dar. Maximal können 9 Slave-Geräte an einer Master-Einheit betrieben werden, wobei jedes Gerät Master sein kann. Maximale Gesamtlänge aller anzubindenden Geräte 50 m. Es können 1-phasige und 3-phasige Geräte angeschlossen werden.

RTT = Rittal TopTherm Kühlgeräte

X1 = Netzanschluss/Türendschalter/Alarmer

X2 = Master-Slave-Anschluss SUB-D 9-polig

X3 = Serielle Schnittstelle SUB-D 9-polig

St. = Stecker SUB-D 9-polig

Bu. = Buchse SUB-D 9-polig

GB	Description The address
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Description
The address of the Master depends on the number of attached Slave units.
(09 = Master connected with 9 Slave units). The address of the Slaves starts with 1. The 2nd number represents the address of the Slave unit. Max. 9 Slave units can be connected to one master unit. Maximum length of the Master/Slave LAN all over is 50 m. Cooling units with different voltages can be connected in a Master/Slave Network.

R1T = Rittal TopTherm cooling units
X1 = AC power supply/door switch/alarms
X2 = Master Slave connection SUB-D 9-pole
X3 = Serial Interface SUB-D 9-pole
St. = Plug SUB-D 9-pole
Bu. = Jack SUB-D 9-pole

F Description

L'adresse de l'appareil maître dépend du nombre d'appareils asservis qui y sont rattachés (09 = appareil maître connecté à 9 appareils asservis). L'adresse d'un appareil asservi commence toujours par le chiffre 1. Le deuxième chiffre représente l'adresse de l'appareil esclave. Chaque appareil peut occuper la fonction d'appareil maître. Et chaque unité maître permet de raccorder 9 unités esclaves au maximum. La longueur totale maximale du réseau maître/esclaves est de 50 m. Il est possible de connecter des appareils de refroidissement de voltages différents (monophasés et triphasés) dans un réseau maître/esclaves.

RTT = Climatiseurs TopTherm de Rittal

X1 = Raccordement au secteur/interrupteur de porte/alarmes

X2 = Raccordement maître/esclave SUB-D 9 pôles

X3 = Interface série SUB-D 9 pôles

St. = Fiche SUB-D 9 pôles

Bu. = Prise SUB-D 9 pôles

NL **Beschrijving**
Het adres van

Beschrijving

Het adres van de master is afhankelijk van het aantal aangesloten slave aggregaten (09 = master met 9 slaves). De adressen van de slave aggregaten beginnen altijd met een 1. Het getal 2 komt overeen met het aantal werkelijke adressen. Maximaal kunnen er 9 slave aggregaten en 1 master geconfigureerd worden, waarbij elke aggregaat de master kan zijn. De maximale lengte van alle gekoppelde aggregaten bedraagt 50 m. Er kunnen 1- en 3-fasen aggregaten worden aangesloten.

RTT = Rittal TopTherm koelaggregaten
X1 = Netaansluiting/deurschakelaar/Alarm
X2 = Master-slave aansluiting SUB-D 9-polig
X3 = Seriële interface SUB-D 9-polig
St. = Connector SUB-D 9-polig
Bu. = Bus SUB-D 9-polig

Beskrivning

Beskrivning
Master-aggregatets adress är beroende på antalet anslutna Slave-aggregat
(09 = Master med 9 Slave-agg.). Slave-aggregatets adress börjar alltid med 1.
Den 2 siffran visar slave-aggregatens adress.
Maximalt 9 Slave-aggregat kan kopplas upp till en master, varvid varje aggregat kan
vara master. Maximal längd av denna Master-Slave LAN är 50 m. Aggregat med
olika spänningar kan kopplas upp i nätet.
RTT = Rittal TopTherm kylaggregat
X0 = Nätanslutning/dörrströmbrytare/larm
X2 = Master-Slave anslutning 9-polig D-Sub
X3 = Seriellt gränssnitt 9-polig D-Sub
St. = Stiftsdon 9-polig D-Sub
Bu. = Hvlsdon 9-polig D-Sub

I **Descrizione**

Indirizzo: L'indirizzo del Master dipende dalla quantità di apparecchi Slave collegati (09 = Master con 9 apparecchi). L'indirizzo di un apparecchio Slave comincia sempre con «1». La seconda cifra corrisponde al vero e proprio indirizzo. È possibile azionare max. 9 apparecchi Slave su di un'unità Master, laddove ci sia un apparecchio Master. La lunghezza massima totale di tutti gli apparecchi collegati è di 50 m. E' possibile collegare apparecchi monofase e trifase.

RTT = Condizionatore Rittal TopTherm

X0 = Allacciamento di rete/Interruttore porta/Allarme

X2 = Connessione Master-Slave SUB-D 9-poli

X3 = Interfaccia seriale SUB-D 9-poli

St = Presa SUB-D 9-poli

Bu = Connettore SUB-D 9-poli

E Descripción
La dirección (o las direcciones) de la actividad.

Descripción:

La dirección del esclavo depende de la cantidad de aparatos esclavos conectados (09 = maestro con 9 aparatos esclavo). La dirección de un aparato esclavo siempre empieza por 1. El número 2 representa la dirección propia. Una unidad maestro no puede gestionar más de 9 aparatos esclavo, cualquier aparato puede ser maestro. La longitud total máxima de todos los aparatos a conectar es de 50 m. Pueden conectarse aparatos de una y tres fases.

RTT = Refrigeradores TopTherm de Rittal

X1 = Conexión a red/Interruptor de puesta/Alarmas

X2 = Conexión maestro-esclavo SUB-D 9 polos

X3 = Interfaz serie SUB-D 9 polos

St. = Conector SUB-D 9 polos

Bu. = Casquillo SUB-D 9 polos

J

説明

マスターユニットのアドレス (Addr.) は、接続されているスレーブユニットの数によって決まります (09 は、40 のスレーブが繋がっている場合です)。スレーブユニットのアドレスは1つ目より続き、2桁目の数字がスレーブユニットを識別する番号になります。1台のマスターユニットに対して、合計9台のスレーブを接続できます。どのユニットをマスターとして選んでもかまいません。ケーブルの長さは、合計で50 m 以内に抑えてください。単相式ユニットと三相式ユニットをつなげることもできます。

RTT = リタラート トップ サーマー クリーニングユニット

X1 = 電源接続、ドアスイッチ、電源出力

X2 = マスター・スレーブ接続 SUB-D 9ピン

X3 = シリアル インターフェース SUB-D 9ピン

St. = プラグ SUB-D 9ピン

Bt. = ソケット SUB-D 9ピン

Verdrahtungsbeispiel: Türeendschalter und Master-Slave-Betrieb

Wiring example: Door limit switch and master/slave operation

Schéma de câblage typique : Interrupteur de porte et régime maître-esclave

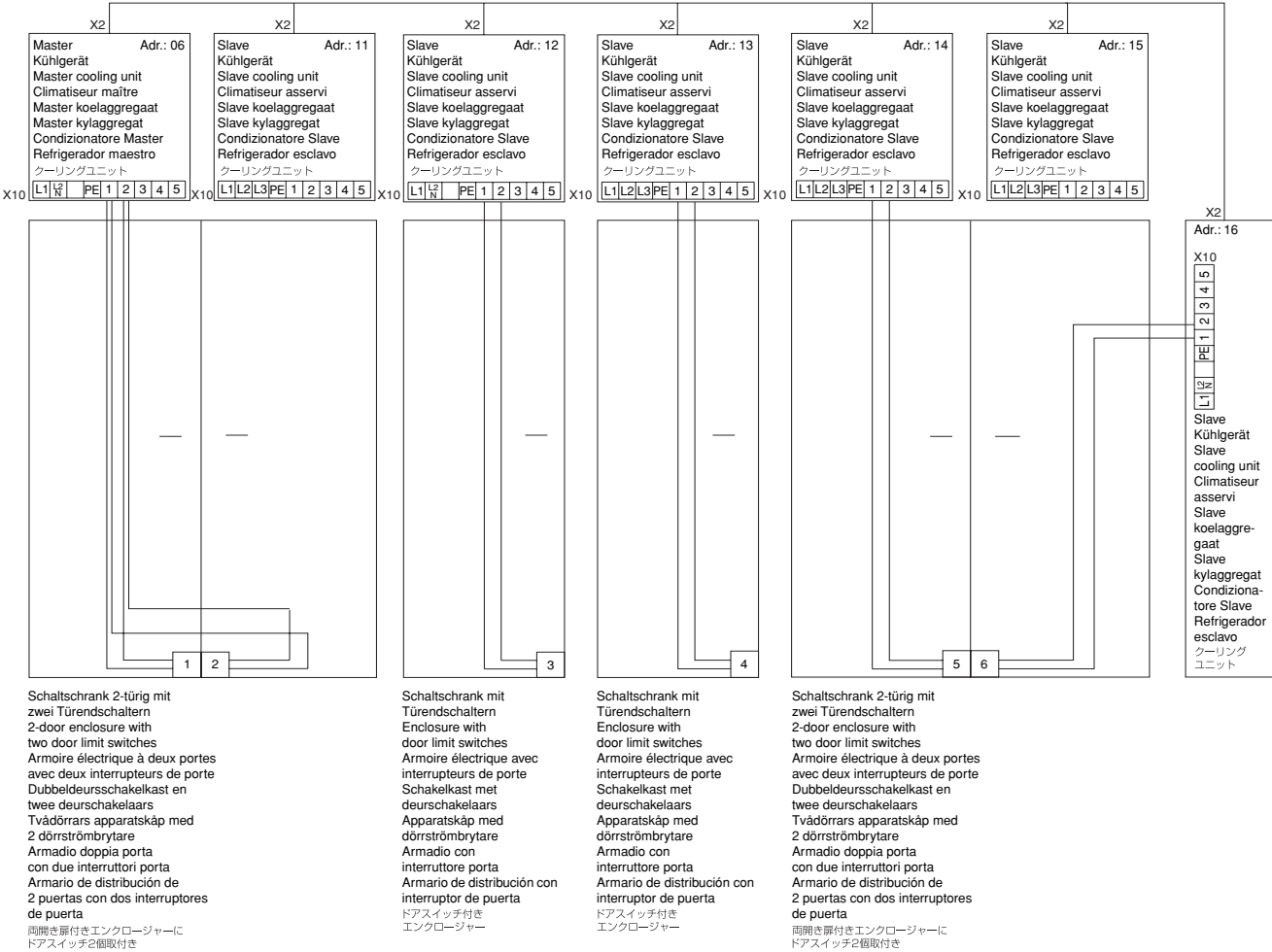
Bekabelingsvoorbeeld: deurschakelaar en master-slave functie

Kopplingsexempel: Dörrströmbrytare och Master-Slavedrift

Esempio di cablaggio: interruttore porta e Funzione Master-Slave

Ejemplo de conexión: Funcionamiento maestro-esclavo

ドアスイッチとマスター・スレーブ運転の配線例



Kennlinienfeld (DIN 3168)

Performance diagram

Diagramme aéraulique

Karakteristiek

Effektdiagram

Diagramma delle potenze

Diagrama de potencia

性能線図

$\dot{Q}_K$

Dauer-Nutzkühlleistung (W)

Cooling output

Puissance frigorifique utilisée

Nuttig koelvermogen

Kyleffekt

Potenza frigorifera utile

Potencia útil de refrigeración

連続有効冷却能力

T_i

Schaltschrank-Innentemperatur (°C)

Enclosure internal temperature

Température intérieure de l'armoire

Temperatuur in de kast

Temperatur inne i skåpet

Temperatura interna dell'armadio

Temperatura interior del armario de mando

内部温度（蒸发器入口温度）

T_u

Umgebungstemperatur (°C)

Ambient temperature

Température ambiante

Omgevingstemperatuur

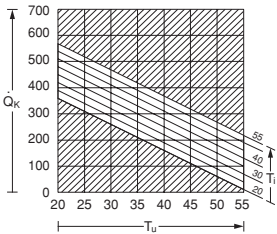
Omgivningstemperatur

Temperatura ambiente

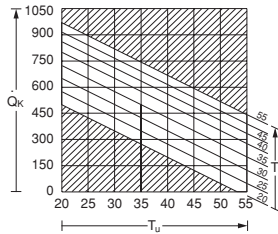
Temperatura ambiente

外部温度（凝縮器入口温度）

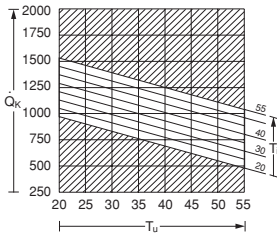
Kennlinienfeld
SK 3302....
(DIN 3168) (50 Hz)



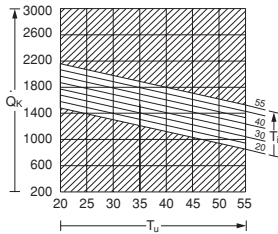
Kennlinienfeld
SK 3303....
(DIN 3168) (50 Hz)



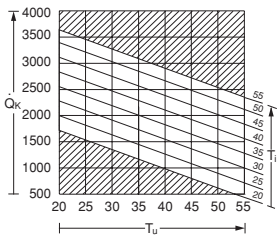
Kennlinienfeld
SK 3304....
(DIN 3168) (50 Hz)



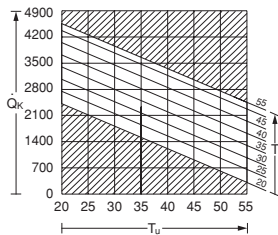
Kennlinienfeld
SK 3305....
(DIN 3168) (50 Hz)



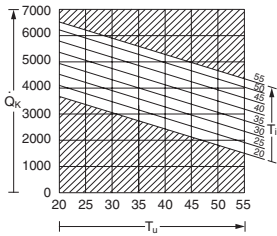
Kennlinienfeld
SK 3328....
(DIN 3168) (50 Hz)



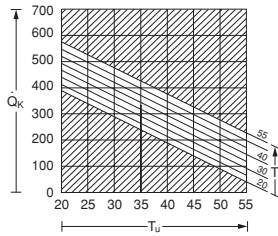
Kennlinienfeld
SK 3329....
(DIN 3168) (50 Hz)



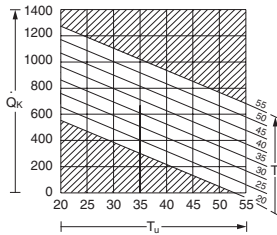
Kennlinienfeld
SK 3332....
(DIN 3168) (50 Hz)



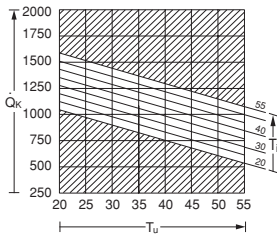
Kennlinienfeld
SK 3302....
(DIN 3168) (60 Hz)



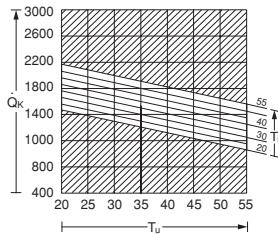
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SK 3303....
(DIN 3168) (60 Hz)



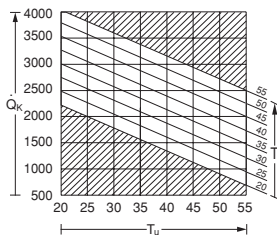
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SK 3304....
(DIN 3168) (60 Hz)



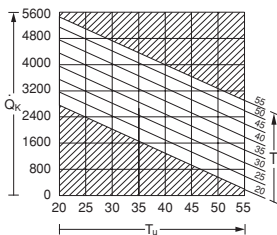
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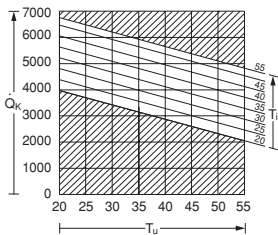
Kennlinienfeld
SK 3328....
(DIN 3168) (60 Hz)



Kennlinienfeld
SK 3329....
(DIN 3168) (60 Hz)



Kennlinienfeld
SK 3332....
(DIN 3168) (60 Hz)





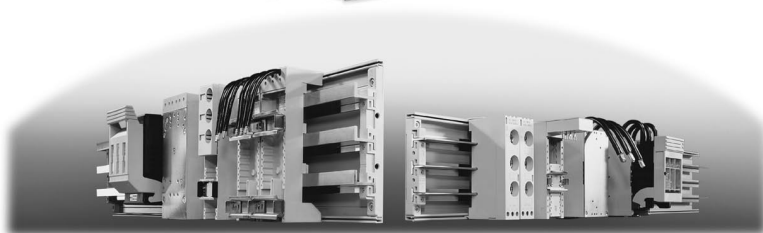
Schaltschrank-Systeme
Industrial Enclosures
Systèmes d'armoires électriques
Schakelkastsystemen
Apparatskåpssystem
Sistemi di armadi per quadri di comando
Sistemas de armarios de distribución
エンクロージャシステム



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Electronic Packaging EL
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Sistemi di allestimento per l'elettronica industriale EL
Sistemas para electrónica EL
エレクトロニクスシステム EL



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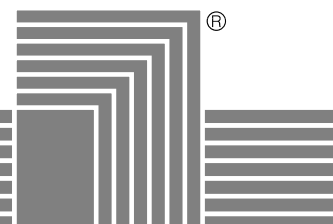


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Umschalten auf Perfektion

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